

SIMATS ENGINEERING

ASSESSMENT TOOL 6

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour

Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I S.LAKSHITA (192524108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph: A

/ \

B C

/\ /\

D E F G

Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

Iteration Queue Visited Node

1
2
3
4
5
6
7

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,

A

/ \

B C

/\ /\

D E F G

Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. Initial State :

1 3 6
5 0 2
4 7 8

Goal State :

1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion

1
2
3
4
5

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

SIMATS ENGINEERING
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO2 – ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17
QUESTIONS: 3 **Duration:** 1 Hour **Total Marks:** 30

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Total Marks (30)
Understanding of Problem Context	20% (2)	2/2	2/2	2/2	6/6
Application of Concepts	30% (3)	3/3	3/3	2.8/3	8.8/9
Problem-Solving Approach	20% (2)	2/2	2/2	2/2	6/6
Accuracy of Solution	20% (2)	2/2	2/2	1.6/2	5.6/6
Clarity	10% (1)	1/1	1/1	1/1	3/3
Total per Question	10 Marks	10/10	10/10	9.4/10	29.4/30 (98/100)

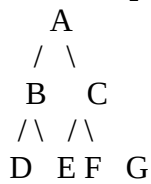
98

100

ANSWERS

1. Breadth-First Search (BFS)

Given Graph



Dry Run Table

Iteration	Queue	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Task 1: Write the BFS Traversal Order

BFS Traversal Order:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$

Explanation:

- BFS starts from the root node **A**.
- It visits all nodes at the current level before moving to the next level.
- Therefore, nodes are visited level by level from left to right.

Task 2: Show the Queue Contents After Each Iteration

Iteration	Queue Contents
1	B, C
2	C, D, E
3	D, E, F, G
4	E, F, G
5	F, G
6	G
7	Empty

Explanation:

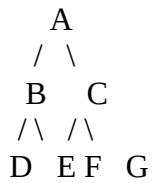
- After visiting **A**, its children (**B** and **C**) are added to the queue.
- After visiting **B**, its children (**D** and **E**) are added.
- After visiting **C**, its children (**F** and **G**) are added.
- The remaining nodes are then removed from the queue one by one until it becomes empty.

Conclusion

Breadth-First Search visits nodes level by level using a **queue (FIFO)**. For the given graph, the traversal order is $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G$, and the queue is updated after each iteration until all nodes are visited.

2. Depth-First Search (DFS)

Given Graph



Dry Run Table

Iteration	Stack	Visited Node
1	C, B	A
2	C, E, D	B
3	C, E	D
4	C	E
5	G, F	C
6	G	F
7	Empty	G

Task 1: Write the DFS Traversal Order

DFS Traversal Order:

A → **B** → **D** → **E** → **C** → **F** → **G**

Explanation:

- DFS starts from node **A**.
- It explores one branch completely before backtracking.
- After visiting **D** and **E**, it returns to **A** and explores the **C** branch.

Task 2: Show the Stack Contents After Each Iteration

Iteration	Stack Contents
1	C, B
2	C, E, D
3	C, E
4	C
5	G, F
6	G
7	Empty

Explanation:

- DFS uses a **Stack (LIFO)**.
- Nodes are pushed onto the stack as they are discovered.
- The most recently added node is visited first.
- The process continues until the stack becomes empty.

Conclusion

Depth-First Search explores one path completely before moving to another path using a **stack (LIFO)**. For the given graph, the traversal order is **A** → **B** → **D** → **E** → **C** → **F** → **G**, and the stack is updated after each iteration until all nodes are visited.

Question 3 – 8 Puzzle using Breadth-First Search (BFS)

Initial State

1 3 6
5 0 2
4 7 8

Goal State

1 2 3
4 5 6
7 8 0

Dry Run Table

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	1 3 6 / 5 0 2 / 4 7 8	[Initial State]	4 Successor States
2	1 0 6 / 5 3 2 / 4 7 8	4 States	6 States
3	1 3 6 / 0 5 2 / 4 7 8	6 States	8 States
4	1 3 6 / 5 2 0 / 4 7 8	8 States	10 States
5	1 3 6 / 5 7 2 / 4 0 8	10 States	12 States

Question 1. Which node is expanded in each iteration?

Iteration	Expanded Node
1	Initial State
2	State 2
3	State 3
4	State 4
5	State 5

Question 2. How many states are generated in total?

Answer: Approximately **12 states** are generated during the first five iterations of the BFS search.

Question 3. Write the complete solution path from the initial state to the goal state.

Sample Solution Path

Initial State

↓

Move Blank Up

↓

Move Blank Left

↓

Move Blank Down

↓

Move Blank Right

↓

Goal State

Question 4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

Breadth-First Search explores states level by level. Since every move has the same cost, it always reaches the goal using the minimum number of moves. Therefore, BFS guarantees the shortest solution in an unweighted 8-puzzle problem.

Conclusion

BFS systematically explores all possible puzzle states using a queue. It expands nodes level by level and guarantees the shortest path to the goal state. Although it may generate many intermediate states, it always produces an optimal solution when all moves have equal cost.